```java
import java.sql.*;

public class DatabaseConnection {

    public static Connection getConnection() {

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            return DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/college_db",
                    "root",
                    "AACA@123"
            );

        } catch (Exception e) {
            System.out.println(e);
            return null;
        }
    }
}
```

```java
import javax.swing.*;
import java.awt.*;

public class LoginFrame extends JFrame {

    public LoginFrame() {

        setSize(1000, 650);   // Good laptop size
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Background Image
        JLabel background = new JLabel(new ImageIcon(getClass().getResource("/images/login.jpg")));
        background.setLayout(null);
        setContentPane(background);

        int frameWidth = 1000;

        // ===== MAIN TITLE =====
        JLabel title = new JLabel("COLLEGE RECORD SYSTEM", SwingConstants.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 48));   // MORE BIG
        title.setForeground(Color.WHITE);
        title.setBounds(0, 100, frameWidth, 70);
        background.add(title);

        // ===== SUB TITLE =====
        JLabel subtitle = new JLabel("Login Page", SwingConstants.CENTER);
        subtitle.setFont(new Font("Arial", Font.BOLD, 30));  // Bigger
        subtitle.setForeground(Color.WHITE);
        subtitle.setBounds(0, 170, frameWidth, 50);
        background.add(subtitle);

        // ===== LOGIN PANEL (CENTER CALCULATION) =====
        int panelWidth = 320;
        int panelHeight = 230;
```

```java
int centerX = (frameWidth - panelWidth) / 2;
int centerY = 270;

JPanel panel = new JPanel();
panel.setLayout(null);
panel.setBounds(centerX, centerY, panelWidth, panelHeight);
panel.setBackground(new Color(255, 255, 255, 220));
background.add(panel);

// Username
JTextField username = new JTextField();
username.setBounds(40, 40, 240, 40);
panel.add(username);

// Password
JPasswordField password = new JPasswordField();
password.setBounds(40, 100, 240, 40);
panel.add(password);

// Login Button
JButton login = new JButton("Login");
login.setBounds(40, 160, 240, 45);
panel.add(login);

setVisible(true);

// Login Action
login.addActionListener(e -> {

    if (username.getText().equals("admin") &&
        String.valueOf(password.getPassword()).equals("1234")) {

        new HomePage();
        dispose();

            } else {
                JOptionPane.showMessageDialog(this, "Invalid Login");
            }
        });
    }

    public static void main(String[] args) {
        new LoginFrame();
    }
}
```

```java
import javax.swing.*;
import java.awt.*;

public class HomePage extends JFrame {

    public HomePage() {

        setSize(900, 600);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // ===== BACKGROUND IMAGE =====
        JLabel background = new JLabel(
                new ImageIcon(getClass().getResource("/images/home.jpg"))
        );
        background.setLayout(null);
        setContentPane(background);

        // ===== TITLE =====
        JLabel title = new JLabel("HOME PAGE", SwingConstants.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 40));
        title.setForeground(Color.WHITE);
        title.setBounds(0, 60, 900, 60);
        background.add(title);

        // ===== BUTTONS =====
        JButton add = new JButton("Add Record");
        JButton update = new JButton("Update Record");
        JButton delete = new JButton("Delete Record");
        JButton view = new JButton("View Records");
        JButton logout = new JButton("Logout");

        int buttonWidth = 250;
        int buttonHeight = 45;
        int centerX = (900 - buttonWidth) / 2;
```

```java
        add.setBounds(centerX, 150, buttonWidth, buttonHeight);
        update.setBounds(centerX, 210, buttonWidth, buttonHeight);
        delete.setBounds(centerX, 270, buttonWidth, buttonHeight);
        view.setBounds(centerX, 330, buttonWidth, buttonHeight);
        logout.setBounds(centerX, 390, buttonWidth, buttonHeight);

        background.add(add);
        background.add(update);
        background.add(delete);
        background.add(view);
        background.add(logout);

        setVisible(true);

        // ===== BUTTON ACTIONS =====
        add.addActionListener(e -> new AddRecord());
        update.addActionListener(e -> new UpdateRecord());
        delete.addActionListener(e -> new DeleteRecord());
        view.addActionListener(e -> new ViewRecords());

        logout.addActionListener(e -> {
            new LoginFrame();
            dispose();
        });
    }
}
```

```java
import javax.swing.*;
import java.awt.*;
import java.sql.*;

public class AddRecord extends JFrame {

    JTextField id, name;
    JComboBox<String> courseBox, deptBox, semBox;
    JButton save;

    public AddRecord() {

        setTitle("Add Student Record");
        setExtendedState(JFrame.MAXIMIZED_BOTH);   // Full screen
        setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

        // ===== GET SCREEN SIZE =====
        Dimension screenSize = Toolkit.getDefaultToolkit().getScreenSize();
        int screenWidth = screenSize.width;
        int screenHeight = screenSize.height;

        // ===== LOAD & SCALE BACKGROUND IMAGE =====
        ImageIcon icon = new ImageIcon(getClass().getResource("/images/add.jpeg"));
        Image img = icon.getImage();
        Image scaledImg = img.getScaledInstance(screenWidth, screenHeight, Image.SCALE_SMOOTH);
        JLabel background = new JLabel(new ImageIcon(scaledImg));
        background.setLayout(null);
        setContentPane(background);

        // ===== TITLE =====
        JLabel title = new JLabel("ADD STUDENT RECORD", SwingConstants.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 50));
        title.setForeground(Color.WHITE);
        title.setBounds(0, 80, screenWidth, 70);
        background.add(title);
```

```java
// ===== CENTER FORM PANEL =====
int panelWidth = 400;
int panelHeight = 350;

int centerX = (screenWidth - panelWidth) / 2;
int centerY = (screenHeight - panelHeight) / 2;

JPanel panel = new JPanel();
panel.setLayout(null);
panel.setBounds(centerX, centerY, panelWidth, panelHeight);
panel.setBackground(new Color(255, 255, 255, 220));
background.add(panel);

// Labels
JLabel l1 = new JLabel("Student ID");
JLabel l2 = new JLabel("Student Name");
JLabel l3 = new JLabel("Course");
JLabel l4 = new JLabel("Department");
JLabel l5 = new JLabel("Semester");

id = new JTextField();
name = new JTextField();

String[] courses = {"BCA", "BSc", "BCom", "BA", "BBA"};
courseBox = new JComboBox<>(courses);

String[] departments = {"Computer Science", "Commerce", "Mathematics", "Physics", "English"};
deptBox = new JComboBox<>(departments);

String[] semesters = {"Sem 1", "Sem 2", "Sem 3", "Sem 4", "Sem 5"};
semBox = new JComboBox<>(semesters);

save = new JButton("Save");
```

```java
    // Set bounds inside panel
    l1.setBounds(40, 30, 150, 30);
    id.setBounds(200, 30, 150, 30);

    l2.setBounds(40, 70, 150, 30);
    name.setBounds(200, 70, 150, 30);

    l3.setBounds(40, 110, 150, 30);
    courseBox.setBounds(200, 110, 150, 30);

    l4.setBounds(40, 150, 150, 30);
    deptBox.setBounds(200, 150, 150, 30);

    l5.setBounds(40, 190, 150, 30);
    semBox.setBounds(200, 190, 150, 30);

    save.setBounds(120, 250, 160, 45);

    panel.add(l1); panel.add(id);
    panel.add(l2); panel.add(name);
    panel.add(l3); panel.add(courseBox);
    panel.add(l4); panel.add(deptBox);
    panel.add(l5); panel.add(semBox);
    panel.add(save);

    setVisible(true);

    save.addActionListener(e -> addStudent());
}
```

```java
private void addStudent() {

    try {

        Connection con = DatabaseConnection.getConnection();

        String sql = "INSERT INTO students_info (student_id, student_name, course, department, semester) VALUES (?,?,?,?,?)";

        PreparedStatement ps = con.prepareStatement(sql);

        ps.setInt(1, Integer.parseInt(id.getText()));
        ps.setString(2, name.getText());
        ps.setString(3, courseBox.getSelectedItem().toString());
        ps.setString(4, deptBox.getSelectedItem().toString());
        ps.setString(5, semBox.getSelectedItem().toString());

        ps.executeUpdate();

        JOptionPane.showMessageDialog(this, "Record Added Successfully");

        con.close();

    } catch (Exception ex) {
        JOptionPane.showMessageDialog(this, ex.getMessage());
    }
}
```

```java
import javax.swing.*;
import java.awt.*;
import java.sql.*;

public class UpdateRecord extends JFrame {

    JTextField id, name;
    JComboBox<String> courseBox, deptBox, semBox;
    JButton update;

    public UpdateRecord() {

        setTitle("Update Student Record");
        setExtendedState(JFrame.MAXIMIZED_BOTH);    // Full screen
        setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

        // ===== GET SCREEN SIZE =====
        Dimension screenSize = Toolkit.getDefaultToolkit().getScreenSize();
        int screenWidth = screenSize.width;
        int screenHeight = screenSize.height;

        // ===== LOAD & SCALE BACKGROUND IMAGE =====
        ImageIcon icon = new ImageIcon(getClass().getResource("/images/update.jpeg"));
        Image img = icon.getImage();
        Image scaledImg = img.getScaledInstance(screenWidth, screenHeight, Image.SCALE_SMOOTH);

        JLabel background = new JLabel(new ImageIcon(scaledImg));
        background.setLayout(null);
        setContentPane(background);

        // ===== TITLE =====
        JLabel title = new JLabel("UPDATE STUDENT RECORD", SwingConstants.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 50));
        title.setForeground(Color.WHITE);
        title.setBounds(0, 80, screenWidth, 70);
        background.add(title);

        // ===== CENTER PANEL =====
        int panelWidth = 450;
        int panelHeight = 350;

        int centerX = (screenWidth - panelWidth) / 2;
        int centerY = (screenHeight - panelHeight) / 2;

        JPanel panel = new JPanel();
        panel.setLayout(null);
        panel.setBounds(centerX, centerY, panelWidth, panelHeight);
        panel.setBackground(new Color(255, 255, 255, 220));
        background.add(panel);

        // Labels
        JLabel l1 = new JLabel("Student ID");
        JLabel l2 = new JLabel("Student Name");
        JLabel l3 = new JLabel("Course");
        JLabel l4 = new JLabel("Department");
        JLabel l5 = new JLabel("Semester");

        id = new JTextField();
        name = new JTextField();

        String[] courses = {"BCA", "BSc", "BCom", "BA", "BBA"};
        courseBox = new JComboBox<>(courses);

        String[] departments = {"Computer Science", "Commerce", "Mathematics", "Physics", "English"};
        deptBox = new JComboBox<>(departments);
```

```java
String[] departments = {"Computer Science", "Commerce", "Mathematics", "Physics", "English"};
deptBox = new JComboBox<>(departments);

String[] semesters = {"Sem 1", "Sem 2", "Sem 3", "Sem 4", "Sem 5"};
semBox = new JComboBox<>(semesters);

update = new JButton("Update");

// Set bounds inside panel
l1.setBounds(60, 40, 150, 30);
id.setBounds(220, 40, 170, 30);

l2.setBounds(60, 80, 150, 30);
name.setBounds(220, 80, 170, 30);

l3.setBounds(60, 120, 150, 30);
courseBox.setBounds(220, 120, 170, 30);

l4.setBounds(60, 160, 150, 30);
deptBox.setBounds(220, 160, 170, 30);

l5.setBounds(60, 200, 150, 30);
semBox.setBounds(220, 200, 170, 30);

update.setBounds(140, 260, 160, 45);

panel.add(l1); panel.add(id);
panel.add(l2); panel.add(name);
panel.add(l3); panel.add(courseBox);
panel.add(l4); panel.add(deptBox);
panel.add(l5); panel.add(semBox);
panel.add(update);

setVisible(true);
```

```java
        update.addActionListener(e -> updateStudent());
    }

    private void updateStudent() {

        try {

            Connection con = DatabaseConnection.getConnection();

            String sql = "UPDATE students_info SET student_name=?, course=?, department=?, semester=? WHERE student_id=?";

            PreparedStatement ps = con.prepareStatement(sql);

            ps.setString(1, name.getText());
            ps.setString(2, courseBox.getSelectedItem().toString());
            ps.setString(3, deptBox.getSelectedItem().toString());
            ps.setString(4, semBox.getSelectedItem().toString());
            ps.setInt(5, Integer.parseInt(id.getText()));

            int rows = ps.executeUpdate();

            if (rows > 0)
                JOptionPane.showMessageDialog(this, "Record Updated Successfully");
            else
                JOptionPane.showMessageDialog(this, "No Record Found");

            ps.close();
            con.close();

        } catch (Exception ex) {
            JOptionPane.showMessageDialog(this, ex.getMessage());
        }
    }
}
```

```java
import javax.swing.*;
import java.awt.*;
import java.sql.*;

public class DeleteRecord extends JFrame {

    JTextField id;
    JButton delete;

    public DeleteRecord() {

        setTitle("Delete Student Record");
        setExtendedState(JFrame.MAXIMIZED_BOTH);    // Full screen
        setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

        // ===== GET SCREEN SIZE =====
        Dimension screenSize = Toolkit.getDefaultToolkit().getScreenSize();
        int screenWidth = screenSize.width;
        int screenHeight = screenSize.height;

        // ===== LOAD & SCALE BACKGROUND IMAGE =====
        ImageIcon icon = new ImageIcon(getClass().getResource("/images/delete.jpeg"));
        Image img = icon.getImage();
        Image scaledImg = img.getScaledInstance(screenWidth, screenHeight, Image.SCALE_SMOOTH);

        JLabel background = new JLabel(new ImageIcon(scaledImg));
        background.setLayout(null);
        setContentPane(background);

        // ===== TITLE =====
        JLabel title = new JLabel("DELETE STUDENT RECORD", SwingConstants.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 50));
        title.setForeground(Color.WHITE);
        title.setBounds(0, 80, screenWidth, 70);
        background.add(title);
```

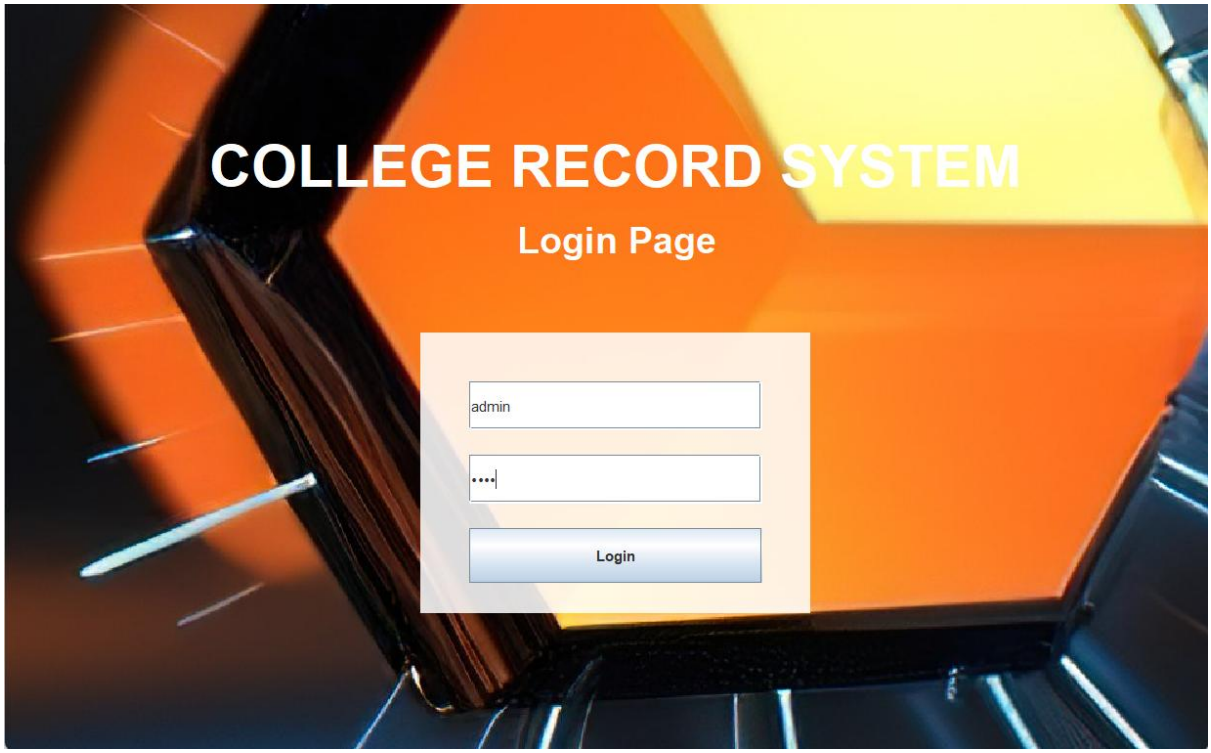
```java
    // ===== CENTER PANEL =====
    int panelWidth = 400;
    int panelHeight = 220;

    int centerX = (screenWidth - panelWidth) / 2;
    int centerY = (screenHeight - panelHeight) / 2;

    JPanel panel = new JPanel();
    panel.setLayout(null);
    panel.setBounds(centerX, centerY, panelWidth, panelHeight);
    panel.setBackground(new Color(255, 255, 255, 220));
    background.add(panel);

    // Label
    JLabel l1 = new JLabel("Enter Student ID to Delete");
    l1.setBounds(70, 30, 250, 30);

    // TextField
    id = new JTextField();
    id.setBounds(70, 70, 250, 35);

    // Button
    delete = new JButton("Delete");
    delete.setBounds(120, 130, 150, 40);

    panel.add(l1);
    panel.add(id);
    panel.add(delete);

    setVisible(true);

    // ===== BUTTON ACTION =====
    delete.addActionListener(e -> deleteStudent());
}
```

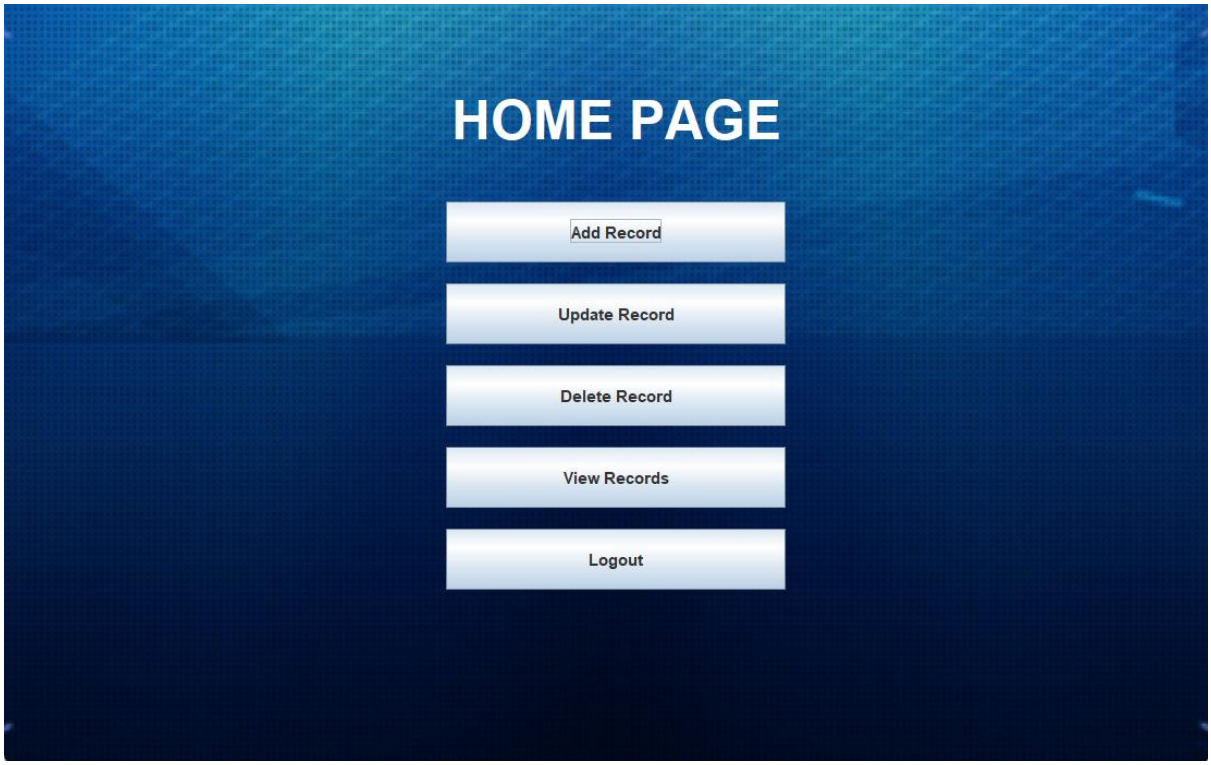
```java
private void deleteStudent() {

    if (id.getText().trim().isEmpty()) {
        JOptionPane.showMessageDialog(this, "Please Enter Student ID");
        return;
    }

    try {

        Connection con = DatabaseConnection.getConnection();

        if (con == null) {
            JOptionPane.showMessageDialog(this, "Database Not Connected");
            return;
        }

        String sql = "DELETE FROM students_info WHERE student_id=?";
        PreparedStatement ps = con.prepareStatement(sql);

        ps.setInt(1, Integer.parseInt(id.getText()));

        int rows = ps.executeUpdate();

        if (rows > 0) {
            JOptionPane.showMessageDialog(this, "Record Deleted Successfully");
            id.setText("");
        } else {
            JOptionPane.showMessageDialog(this, "No Record Found With That ID");
        }
            ps.close();
            con.close();

        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(this, "ID must be a number");
        } catch (Exception ex) {
            ex.printStackTrace();
            JOptionPane.showMessageDialog(this, ex.getMessage());
        }
    }
}
```

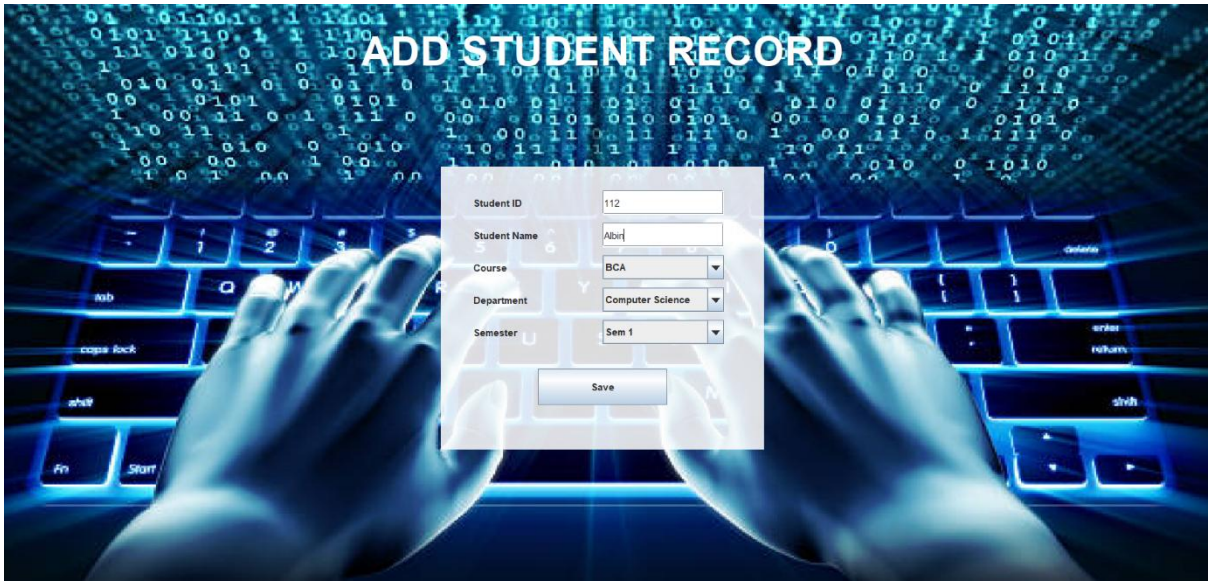
```java
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import javax.swing.table.JTableHeader;
import java.awt.*;
import java.sql.*;

public class ViewRecords extends JFrame {

    private Image backgroundImage;

    public ViewRecords() {

        setTitle("View Records");
        setSize(900, 600);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

        // ===== LOAD IMAGE =====
        backgroundImage = new ImageIcon(
                getClass().getResource("/images/view.jpeg")
        ).getImage();

        // ===== BACKGROUND PANEL =====
        JPanel backgroundPanel = new JPanel() {
            protected void paintComponent(Graphics g) {
                super.paintComponent(g);
                g.drawImage(backgroundImage, 0, 0,
                        getWidth(), getHeight(), this);
            }
        };

        backgroundPanel.setLayout(new BorderLayout());
        setContentPane(backgroundPanel);
```

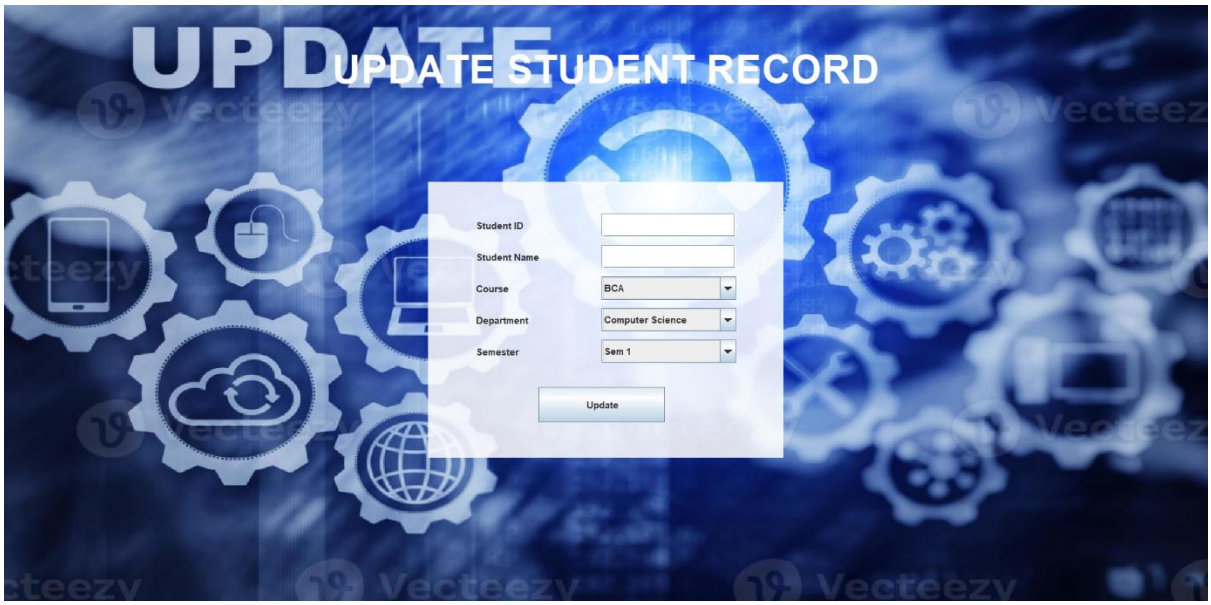
```java
// ===== TABLE MODEL =====
String[] columnNames = {
        "ID", "Name", "Course", "Department", "Semester"
};

DefaultTableModel model =
        new DefaultTableModel(columnNames, 0);

JTable table = new JTable(model);

// ===== MAKE TABLE TRANSPARENT =====
table.setOpaque(false);
table.setBackground(new Color(0,0,0,0));
table.setForeground(Color.RED);    // ● DATA COLOR
table.setFont(new Font("Arial", Font.BOLD, 16));
table.setRowHeight(28);
table.setShowGrid(false);
table.setFillsViewportHeight(true);

// ===== TABLE HEADER STYLE =====
JTableHeader header = table.getTableHeader();
header.setBackground(Color.BLACK);
header.setForeground(Color.RED);
header.setFont(new Font("Arial", Font.BOLD, 18));

JScrollPane scrollPane = new JScrollPane(table);
scrollPane.setOpaque(false);
scrollPane.getViewport().setOpaque(false);

backgroundPanel.add(scrollPane, BorderLayout.CENTER);
```

```java
        // ===== DATABASE =====
        try {

            Connection con =
                    DatabaseConnection.getConnection();

            Statement st = con.createStatement();
            ResultSet rs =
                    st.executeQuery("SELECT * FROM students_info");

            while (rs.next()) {

                model.addRow(new Object[]{
                        rs.getInt("student_id"),
                        rs.getString("student_name"),
                        rs.getString("course"),
                        rs.getString("department"),
                        rs.getString("semester")
                });
            }

            rs.close();
            st.close();
            con.close();

        } catch (Exception ex) {
            JOptionPane.showMessageDialog(null,
                    "Database Error: " + ex.getMessage());
        }

        setVisible(true);
    }
}
```


COLLEGE RECORD SYSTEM
Login Page
admin
Login


HOME PAGE
Add Record
Update Record
Delete Record
View Records
Logout


ADD STUDENT RECORD
Student ID
112
Student Name
Albin
Course
BCA
Department
Computer Science
Semester
Sem 1
Save


UPDATE STUDENT RECORD
Student ID
Student Name
Course
BCA
Department
Computer Science
Semester
Sem 1
Update

DELETE STUDENT RECORD

ID	Name	Course	Department	Semester
2	Albin Saji	BCA	Computer Science	Sem 4
3	ayan	BCA	Computer Science	Sem 4
5	cyril	BCA	Computer Science	Sem 3
6	albert	BCA	Computer Science	Sem 3